Year 11 Higher

REVISION PACK

Progress Check 2

Wednesday 21 October 2026

Name: _____

Class: _____

Units in this pack

- Graphs 5
- Handling Data 4
- Shape and Space 5

Graphs 5 · Simultaneous equations (Lesson 3.1)

Worked example: adding to remove a letter

Solve the simultaneous equations

$2x + 3y = 12$

$4x - 3y = 6$

Step 1	$+3y$ and $-3y$: add	$6x = 18$
Step 2	Solve	$x = 3$
Step 3	Put x in	$2(3) + 3y = 12$
Step 4	Solve	$y = 2$
Step 5	Answer	$x = 3,\ y = 2$

Practice: adding to remove a letter

1.1 Solve the simultaneous equations

$4x + 3y = 18$

$2x - 3y = 0$

1.2 Solve the simultaneous equations

$5x + 2y = 18$

$3x - 2y = -2$

1.3 Solve the simultaneous equations

$2x + 5y = 13$

$4x - 5y = 11$

1.4 Solve the simultaneous equations

$3x + 2y = 18$

$x - 2y = -2$

1.5 Solve the simultaneous equations

$6x + y = 11$

$2x - y = -3$

1.6 Solve the simultaneous equations

$5x + 4y = 6$

$3x - 4y = 10$

Graphs 5 · Simultaneous equations (Lesson 3.1)

Worked example: multiplying first

Solve the simultaneous equations

$3x + 2y = 12$

$2x - 3y = -5$

Step 1	First × 3	$9x + 6y = 36$
Step 2	Second × 2	$4x - 6y = -10$
Step 3	Add	$13x = 26$
Step 4	Solve	$x = 2$
Step 5	Put x in	$3(2) + 2y = 12$
Step 6	Solve	$y = 3$

Practice: multiplying first

1.7 Solve the simultaneous equations

$5x + 3y = 12$

$2x - 4y = 10$

1.8 Solve the simultaneous equations

$3x + 4y = 10$

$5x - 3y = 7$

1.9 Solve the simultaneous equations

$2x + 5y = 13$

$3x + 2y = 3$

1.10 Solve the simultaneous equations

$4x + 5y = 13$

$3x - 2y = 4$

1.11 Solve the simultaneous equations

$6x + 5y = 16$

$4x - 3y = -2$

1.12 Solve the simultaneous equations

$3x + 2y = 11$

$4x - 5y = 7$

Graphs 5 · Simultaneous equations (Lesson 3.1)

Worked example: problems with two unknowns

4 pens and 1 ruler cost 2.100 KD. 2 pens and 3 rulers cost 1.800 KD. Work out the cost of one pen and the cost of one ruler.

Step 1	Equation 1	$4p + r = 2.1$
Step 2	Equation 2 × 2	$4p + 6r = 3.6$
Step 3	Subtract	$5r = 1.5$
Step 4	Solve	$r = 0.3$
Step 5	Put r in	$4p + 0.3 = 2.1$
Step 6	Solve	$p = 0.45$
Step 7	Answer	$\text{pen } 0.450 \text{ KD},\ \text{ruler } 0.300 \text{ KD}$

Practice: problems with two unknowns

1.13 2 notebooks and 3 pencils cost 2.250 KD. 1 notebook and 4 pencils cost 1.750 KD. Work out the cost of one notebook and the cost of one pencil.

1.14 5 apples and 2 mangos cost 3.300 KD. 3 apples and 4 mangos cost 3.800 KD. Work out the cost of one apple and the cost of one mango.

1.15 3 coffees and 1 cake cost 4.100 KD. 2 coffees and 5 cakes cost 4.900 KD. Work out the cost of one coffee and the cost of one cake.

1.16 2 sandwiches and 1 juice cost 3.500 KD. 1 sandwich and 3 juices cost 3.000 KD. Work out the cost of one sandwich and the cost of one juice.

1.17 4 pencils and 3 erasers cost 2.800 KD. 2 pencils and 5 erasers cost 3.500 KD. Work out the cost of one pencil and the cost of one eraser.

1.18 3 books and 2 magazines cost 10.000 KD. 1 book and 2 magazines cost 5.000 KD. Work out the cost of one book and the cost of one magazine.

Graphs 5 · Simultaneous equations (Lesson 3.1)

Worked example: adding the equations

Solve the simultaneous equations

$4x + y = 11$

$2x - y = 1$

Step 1	$+y$ and $-y$: add	$6x = 12$
Step 2	Solve	$x = 2$
Step 3	Put x in	$4(2) + y = 11$
Step 4	Solve	$y = 3$

Practice: adding the equations

1.19 Solve the simultaneous equations

$3x + y = 7$

$2x - y = 3$

1.20 Solve the simultaneous equations

$4x + y = 11$

$x - y = -1$

1.21 Solve the simultaneous equations

$2x + y = 8$

$5x - y = 13$

1.22 Solve the simultaneous equations

$x + y = 5$

$3x - y = 11$

1.23 Solve the simultaneous equations

$5x + y = 9$

$2x - y = -2$

1.24 Solve the simultaneous equations

$3x + y = 11$

$4x - y = 3$

Handling Data 4 · Collecting data and averages (Lessons 4.1, 4.5)

Worked example: questionnaires

Ali uses this question on a questionnaire to find out how much students spend on snacks. Write down two things that are wrong with the question.

How much do you spend on snacks?
0–1 KD ☐ 1–2 KD ☐ 2–3 KD ☐

Step 1	Is there a time?	No: per day or per week?
Step 2	Do boxes overlap?	Yes: 1 KD is in two boxes
Step 3	Is every answer there?	No box for more than 3 KD

Practice: questionnaires

2.1 Huda uses this question on a questionnaire to find out how much time students spend on homework. Write down two things that are wrong with the question.

How many hours of homework do you do?
0–2 ☐ 2–4 ☐ 4–6 ☐

2.2 Majed uses this question on a questionnaire to find out how often people exercise. Write down two things that are wrong with the question.

How many times do you exercise?
1–3 ☐ 3–5 ☐ 5–7 ☐

2.3 Layla uses this question on a questionnaire to find out how much people spend on lunch. Write down two things that are wrong with the question.

How much do you spend on lunch?
0–2 KD ☐ 2–4 KD ☐ 4–6 KD ☐

Practice: questionnaires (continued)

2.4 Omar uses this question on a questionnaire to find out how much time people spend on their phones. Write down two things that are wrong with the question.

How many hours do you spend on your phone?
0–2 ☐ 2–4 ☐ 4–6 ☐

2.5 Mona uses this question on a questionnaire to find out how much people spend on clothes. Write down two things that are wrong with the question.

How much do you spend on clothes?
0–10 KD ☐ 10–20 KD ☐ 20–30 KD ☐

2.6 Fahad uses this question on a questionnaire to find out how many books students read. Write down two things that are wrong with the question.

How many books do you read?
1–2 ☐ 2–3 ☐ 3–4 ☐

Handling Data 4 · Collecting data and averages (Lessons 4.1, 4.5)

Worked example: averages from a frequency table

The table shows the numbers of goals scored by a team in 20 matches.

Number of goals	0	1	2	3
Frequency	5	8	4	3

(a) Write down the mode. (b) Work out the mean number of goals.

(a)	Step 1	Highest frequency	8
	Step 2	Answer	1
(b)	Step 1	Goals × frequency	$0 + 8 + 8 + 9 = 25$
	Step 2	Total frequency	$5 + 8 + 4 + 3 = 20$
	Step 3	Mean	$25 \div 20 = 1.25$

Practice: averages from a frequency table

2.7 The table shows the numbers of goals scored by a team in 25 matches.

Number of goals	0	1	2	3	4
Frequency	3	7	8	5	2

(a) Write down the mode. (b) Work out the mean.

2.8 The table shows the numbers of emails Sara got on 25 days.

Number of emails	1	2	3	4	5
Frequency	2	9	6	4	4

(a) Write down the mode. (b) Work out the mean.

2.9 The table shows the numbers of visitors to a shop each hour for 25 hours.

Number of visitors	0	1	2	3	4
Frequency	6	5	4	8	2

(a) Write down the mode. (b) Work out the mean.

2.10 The table shows the numbers of pets owned by 20 students.

Number of pets	0	1	2	3	4
Frequency	5	9	4	1	1

(a) Write down the mode. (b) Work out the mean.

2.11 The table shows the numbers of books read by 20 students.

Number of books	0	1	2	3	4
Frequency	2	4	8	4	2

(a) Write down the mode. (b) Work out the mean.

2.12 The table shows the numbers of brothers and sisters of 20 students.

Number of brothers and sisters	0	1	2	3	4
Frequency	3	8	6	2	1

(a) Write down the mode. (b) Work out the mean.

Handling Data 4 · Collecting data and averages (Lessons 4.1, 4.5)

Worked example: a missing number from the mean

The mean of four numbers is 9. Three of the numbers are 7, 8 and 12. Work out the fourth number.

Step 1	Total	$4 \times 9 = 36$
Step 2	Add the three	$7 + 8 + 12 = 27$
Step 3	Subtract	$36 - 27 = 9$

Practice: a missing number from the mean

2.13 The mean of five numbers is 16. Four of the numbers are 12, 14, 19 and 20. Work out the fifth number.

2.14 The mean of six numbers is 11. Five of the numbers are 8, 9, 13, 14 and 10. Work out the sixth number.

2.15 The mean of five numbers is 23. Four of the numbers are 18, 25, 27 and 20. Work out the fifth number.

2.16 The mean of five numbers is 12. Four of the numbers are 9, 11, 14 and 15. Work out the fifth number.

2.17 The mean of four numbers is 20. Three of the numbers are 18, 25 and 16. Work out the fourth number.

2.18 The mean of six numbers is 15. Five of the numbers are 12, 18, 14, 20 and 11. Work out the sixth number.

Shape and Space 5 · Circle theorems (Lessons 5.1, 5.2, 5.3)

Worked example: the angle in a semicircle

A, B and C are points on a circle, centre O. AB is a diameter. Angle CAB = 34°. Work out angle ABC, z.
Give a reason for your answer.

Step 1	Angle in a semicircle	$ACB = 90^\circ$
Step 2	Angles in a triangle	$z = 180 - 90 - 34$
Step 3	Answer	$z = 56^\circ$

Practice: the angle in a semicircle

3.1

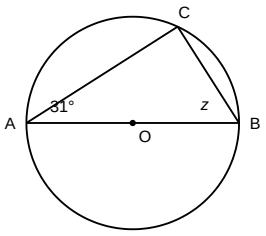


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
AB is a diameter. Angle CAB = 31°
Work out the value of z.
Give a reason for your answer.

3.2

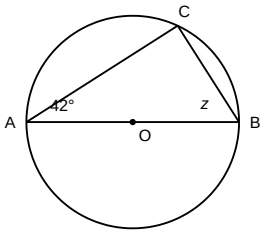


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
AB is a diameter. Angle CAB = 42°
Work out the value of z.
Give a reason for your answer.

3.3

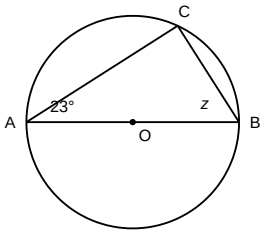


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
AB is a diameter. Angle CAB = 23°
Work out the value of z.
Give a reason for your answer.

Practice: the angle in a semicircle (continued)

3.4

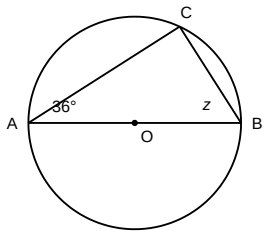


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
AB is a diameter. Angle CAB = 36°
Work out the value of z.
Give a reason for your answer.

3.5

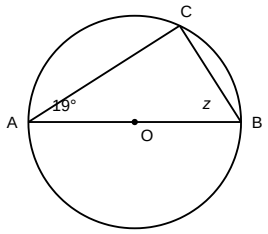


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
AB is a diameter. Angle CAB = 19°
Work out the value of z.
Give a reason for your answer.

3.6

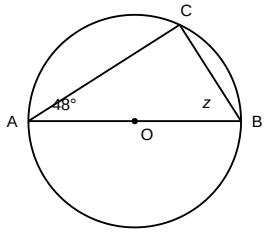


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
AB is a diameter. Angle CAB = 48°
Work out the value of z.
Give a reason for your answer.

Shape and Space 5 · Circle theorems (Lessons 5.1, 5.2, 5.3)

Worked example: angles in the same segment

A, B, C and D are points on a circle. Angle ACB = 44°. Work out angle ADB, w. Give a reason for your answer.

Step 1	Same segment: equal	$w = 44^\circ$
Step 2	Reason	angles in the same segment are equal

Practice: angles in the same segment

3.7

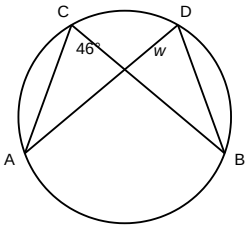


Diagram **NOT** accurately drawn

A, B, C and D are points on a circle.
Angle ACB = 46°
Work out the value of w.
Give a reason for your answer.

3.8

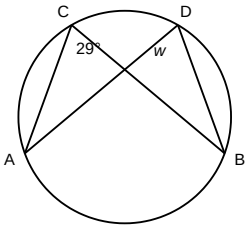


Diagram **NOT** accurately drawn

A, B, C and D are points on a circle.
Angle ACB = 29°
Work out the value of w.
Give a reason for your answer.

3.9

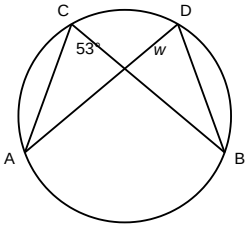
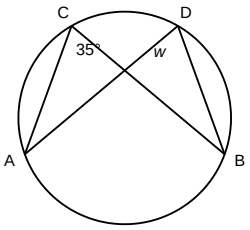


Diagram **NOT** accurately drawn

A, B, C and D are points on a circle.
Angle ACB = 53°
Work out the value of w.
Give a reason for your answer.

Practice: angles in the same segment (continued)

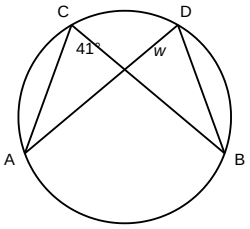
3.10



A, B, C and D are points on a circle.
Angle ACB = 35°
Work out the value of w.
Give a reason for your answer.

Diagram **NOT** accurately drawn

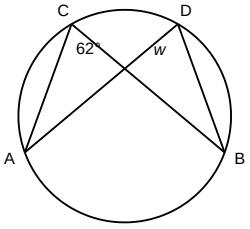
3.11



A, B, C and D are points on a circle.
Angle ACB = 41°
Work out the value of w.
Give a reason for your answer.

Diagram **NOT** accurately drawn

3.12



A, B, C and D are points on a circle.
Angle ACB = 62°
Work out the value of w.
Give a reason for your answer.

Diagram **NOT** accurately drawn

Shape and Space 5 · Circle theorems (Lessons 5.1, 5.2, 5.3)

Worked example: cyclic quadrilaterals

$ABCD$ is a cyclic quadrilateral. Angle ABC = 112°. Work out angle ADC, x. Give a reason for your answer.

Step 1	Opposite angles add to 180°	$x = 180 - 112$
Step 2	Answer	$x = 68^\circ$

Practice: cyclic quadrilaterals

3.13

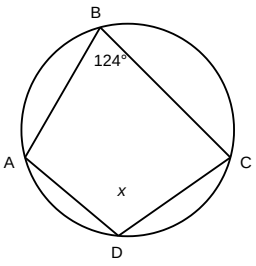


Diagram **NOT** accurately drawn

A, B, C and D are points on a circle.
Angle ABC = 124°
Work out the value of x.
Give a reason for your answer.

3.14

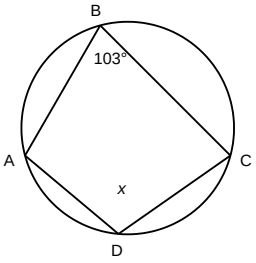


Diagram **NOT** accurately drawn

A, B, C and D are points on a circle.
Angle ABC = 103°
Work out the value of x.
Give a reason for your answer.

3.15

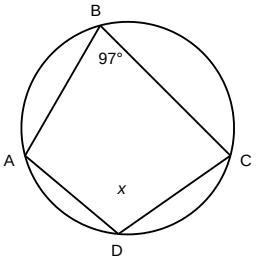


Diagram **NOT** accurately drawn

A, B, C and D are points on a circle.
Angle ABC = 97°
Work out the value of x.
Give a reason for your answer.

Practice: cyclic quadrilaterals (continued)

3.16

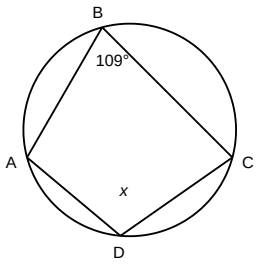


Diagram **NOT** accurately drawn

A, B, C and D are points on a circle.
Angle ABC = 109°
Work out the value of x.
Give a reason for your answer.

3.17

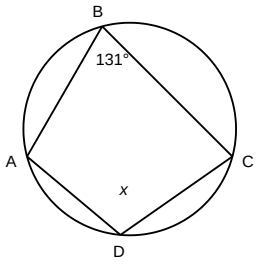


Diagram **NOT** accurately drawn

A, B, C and D are points on a circle.
Angle ABC = 131°
Work out the value of x.
Give a reason for your answer.

3.18

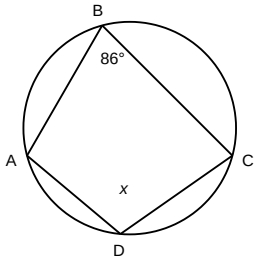


Diagram **NOT** accurately drawn

A, B, C and D are points on a circle.
Angle ABC = 86°
Work out the value of x.
Give a reason for your answer.

Shape and Space 5 · Circle theorems (Lessons 5.1, 5.2, 5.3)

Worked example: the alternate segment theorem

TAS is a tangent to a circle at A. Angle BAS = 64°. Work out angle ACB, y. Give a reason for your answer.

Step 1	Tangent and chord	64°
Step 2	Answer	$y = 64^\circ$
Step 3	Reason	alternate segment theorem

Practice: the alternate segment theorem

3.19

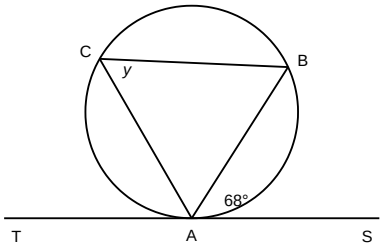


Diagram **NOT** accurately drawn

A, B and C are points on a circle.
TAS is the tangent to the circle at A.
Angle BAS = 68°
Work out the value of y.
Give a reason for your answer.

3.20

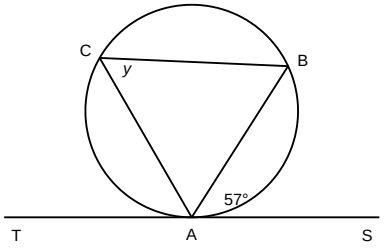


Diagram **NOT** accurately drawn

A, B and C are points on a circle.
TAS is the tangent to the circle at A.
Angle BAS = 57°
Work out the value of y.
Give a reason for your answer.

3.21

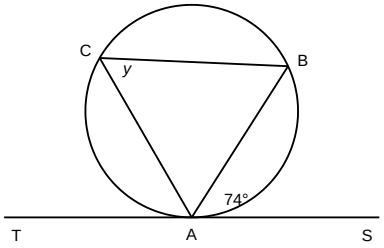


Diagram **NOT** accurately drawn

A, B and C are points on a circle.
TAS is the tangent to the circle at A.
Angle BAS = 74°
Work out the value of y.
Give a reason for your answer.

Practice: the alternate segment theorem (continued)

3.22

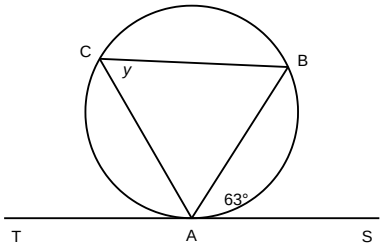


Diagram **NOT** accurately drawn

A, B and C are points on a circle.
TAS is the tangent to the circle at A.
Angle BAS = 63°
Work out the value of y.
Give a reason for your answer.

3.23

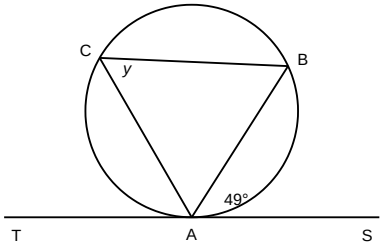


Diagram **NOT** accurately drawn

A, B and C are points on a circle.
TAS is the tangent to the circle at A.
Angle BAS = 49°
Work out the value of y.
Give a reason for your answer.

3.24

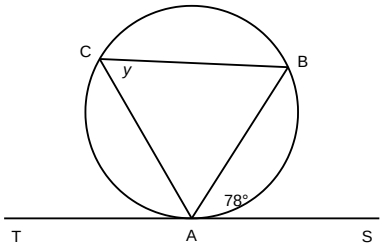


Diagram **NOT** accurately drawn

A, B and C are points on a circle.
TAS is the tangent to the circle at A.
Angle BAS = 78°
Work out the value of y.
Give a reason for your answer.

Shape and Space 5 · Circle theorems (Lessons 5.1, 5.2, 5.3)

Worked example: the angle at the centre

O is the centre of a circle. Angle AOB = 130°. Work out the size of angle ACB. Give a reason for your answer.

Step 1	Centre is twice the edge	$130 \div 2$
Step 2	Answer	65°

Practice: the angle at the centre

3.25

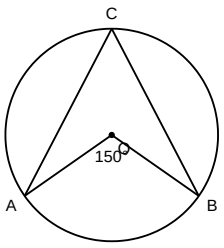


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
Angle AOB = 150°
Work out the size of angle ACB.
Give a reason for your answer.

3.26

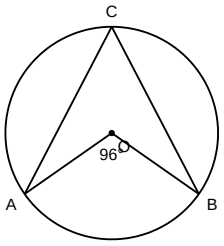


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
Angle AOB = 96°
Work out the size of angle ACB.
Give a reason for your answer.

3.27

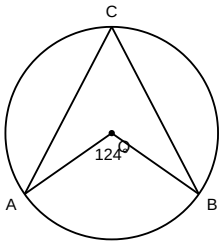


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
Angle AOB = 124°
Work out the size of angle ACB.
Give a reason for your answer.

Practice: the angle at the centre (continued)

3.28

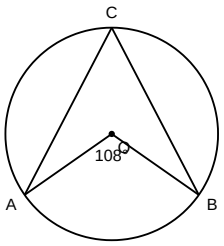


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
Angle AOB = 108°
Work out the size of angle ACB.
Give a reason for your answer.

3.29

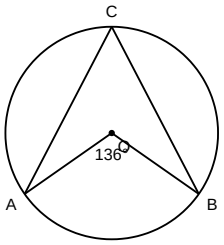


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
Angle AOB = 136°
Work out the size of angle ACB.
Give a reason for your answer.

3.30

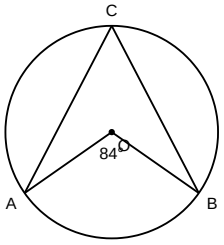


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O.
Angle AOB = 84°
Work out the size of angle ACB.
Give a reason for your answer.